Wentao Zhang
Research Assistant, Power Electronics and Machines Centre
Faculty of Engineering, University of Nottingham
wentao.zhang@nottingham.ac.uk · (+44) 7402 114 543

5 June 2026

Propulsion Team
Heart Aerospace
Los Angeles, California, United States

Dear Hiring Team,

Application for Electric Motor Design Engineer, Propulsion (Job 4013301009)

I am writing to apply for the Electric Motor Design Engineer position on the Propulsion team. I am currently a Research Assistant at the **Power Electronics and Machines Centre (PEMC), University of Nottingham**, a leading research group for transport electrification and electric aircraft propulsion, having completed my **PhD in Electrical Engineering at Southeast University, Nanjing**, in September 2025. My work is the **design, multi-physics optimization, and dynamometer validation of high-speed, high-power-density PM machines**, and the hybrid-electric mission of the ES-30, with the X2 demonstrator's hybrid flight campaign ahead, is the kind of propulsion problem I would like to work on.

The most direct match to the role is hardware. Through a multi-year collaboration with **Inovance**, a world-leading EV traction-machine supplier, I took two oil-cooled **high-voltage PM traction machines** through the **full design-build-test cycle**: a **20,000 rpm, 270 kW** machine (380 Nm, **7.87 kW/kg**, 97.4% peak efficiency) and an 18,000 rpm, 113 kW machine. Each went from electromagnetic, thermal, and mechanical design through **engineering-drawing release and tolerance specification** to **AVL dynamometer characterization** across the full efficiency map. Power density and thermal margin governed every design decision, the same currencies that govern aircraft propulsion motors.

My design record also supports the role's **electric motor topology trade studies**. It covers interior PM, wound-field switched flux (magnet-free), and hybrid-excitation machines within a **multi-objective optimization** framework coupling electromagnetic FEA, lumped-parameter thermal networks, and mechanical-stress evaluation. I work in **JMAG, ANSYS Maxwell, and Motor-CAD** alongside an in-house Python finite-element code package, and at PEMC I am developing operator-learning surrogate models that cut evaluation time by more than 94% relative to direct FEA, which widens the design space a trade study can afford to cover.

On **design for manufacturing**, I developed a systematic *n*-layer flat-wire winding methodology for hairpin PM machines (*IEEE Trans. Transportation Electrification*), covering conductor layout and AC-loss reduction, and co-authored studies on stator-core manufacturing effects such as the vibration behavior of glued versus welded lamination stacks. The prototype programs gave me working knowledge of **stator winding, lamination stacking, and varnish impregnation** processes, and of rotor mechanical-stress design at 20,000 rpm, including stress-relieving groove design.

On **test-stand integration**, beyond the AVL dynamometer campaigns I designed, built, and commissioned a **three-phase full-bridge converter** and its control algorithm for machine prototype testing, and I model **PWM-excitation effects on loss and temperature** so that bench observations close the loop back into design.

I have authored 22 journal papers (17 in IEEE Transactions), 18 conference papers, and 9 patents, and received 1st Prize in the 2025 IEEE TEC Electric Vehicle Design Competition and the 2025 IEEE Transactions on Transportation Electrification Prize Paper Award. My resume is attached, and I would be glad to discuss how I could contribute to the ES-30 propulsion system.

Yours sincerely,

Wentao Zhang